PREVIEW

PAPER BROS

A ONE STOP SOLUTION FOR YOUR BOARDS

DEVELOPED BY STUDENTS, FOR STUDENTS

O LEVEL CHEMISTRY 5070

COMPLETE NOTES · THEORY + PRACTICAL

NOTES + DIAGRAMS + PRACTICE QUESTIONS + ANSWERS

13 CHAPTERS · 161 MCQs · 57 STRUCTURED QUESTIONS · 2026–2028 SYLLABUS



Preview

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How to use these notes

These notes follow the Cambridge O Level Chemistry 5070 syllabus for 2026–2028 topic by topic. Each of Chapters 1–12 matches one syllabus topic: a coloured opener, the teaching notes (every learning objective is listed in a **"You should be able to"** box at the start of each section), a summary, then a practice section in exam style with full answers. Chapter 13 condenses the whole course into the reactions, catalysts, definitions, colour changes and lists you must memorise.

The colour code

DEFINITION
Learn these word-for-word — the mark scheme does.

KEY POINT
The central idea of the section.

EXAM TIP
How to pick up (or avoid dropping) marks.

COMMON MISTAKE
Errors examiners see every year.

BEYOND THE SYLLABUS
Useful extras, shortcuts and formats that deepen understanding.

MEMORY TRICK
Mnemonics to make facts stick.

WORKED EXAMPLE
Step-by-step solutions with units and significant figures.

Boxed formula = must learn

The examination at a glance

Paper	Type	Time	Marks	Weighting
1	Multiple choice — 40 four-option questions	1 h	40	30%
2	Theory — short-answer and structured questions	1 h 45 min	80	50%
3 *or* **4**	Practical Test (lab) *or* Alternative to Practical (written)	1 h 30 min / 1 h	40	20%

All candidates take Papers 1 and 2 plus one of Papers 3 or 4. A copy of the Periodic Table and the qualitative analysis notes are given in the exam — but you should still know them (see Chapters 12 and 13). Calculators are allowed in all papers.

A study method that works

1 · Read
Work through a section, then close the book and write down its key points from memory.

2 · Practise
Do the chapter's MCQs and structured questions under timed conditions.

3 · Mark
Use the mark schemes strictly — note the exact words that score.

4 · Revise
Use Chapter 13 and the summary boxes for spaced revision in the final weeks.

Conventions used

State symbols (s), (l), (g), (aq) · concentrations in mol/dm³ or g/dm³ · 1 dm³ = 1000 cm³ · molar gas volume = 24 dm³ at r.t.p. · ΔH in kJ/mol · Roman numerals show oxidation numbers, e.g. iron(III) · the equilibrium sign is ⇌ · relative atomic masses are taken from the Periodic Table opposite the contents.

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PT The Periodic Table of Elements — 5

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The Periodic Table of Elements

	I	II										III	IV	V	VI	VII	VIII
1	1 H hydrogen 1																2 He helium 4
2	3 Li lithium 7	4 Be beryllium 9										5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20
3	11 Na sodium 23	12 Mg magnesium 24										13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40
4	19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84
5	37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium –	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131
6	55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium –	85 At astatine –	86 Rn radon –
7	87 Fr francium –	88 Ra radium –	89–103 actinoids	104 Rf rutherfordium –	105 Db dubnium –	106 Sg seaborgium –	107 Bh bohrium –	108 Hs hassium –	109 Mt meitnerium –	110 Ds darmstadtium –	111 Rg roentgenium –	112 Cn copernicium –	113 Nh nihonium –	114 Fl flerovium –	115 Mc moscovium –	116 Lv livermorium –	117 Ts tennessine –	118 Og oganesson –

atomic number
X
name
relative atomic mass

lanthanoids	57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium –	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
actinoids	89 Ac actinium –	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium –	94 Pu plutonium –	95 Am americium –	96 Cm curium –	97 Bk berkelium –	98 Cf californium –	99 Es einsteinium –	100 Fm fermium –	101 Md mendelevium –	102 No nobelium –	103 Lr lawrencium –

alkali metals (Group I) · Group II metals · transition elements · other metals · metalloids · other non-metals · halogens (Group VII) · noble gases (Group VIII) · lanthanoids · actinoids

The volume of one mole of any gas is **24 dm³** at room temperature and pressure (r.t.p.). · Avogadro constant = 6.02×10^{23} per mol.

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CHAPTER

01

States of Matter

Ice melts in a drink, a puddle dries on a sunny day, and the smell of cooking drifts from the kitchen to every room. All of these happen because matter is made of tiny particles that never stop moving. This chapter builds the **kinetic particle theory**, the single idea you will use to explain melting, boiling, gas pressure and diffusion, and which comes back again in bonding, rates of reaction and equilibrium.

IN THIS CHAPTER

- 1.1 Solids, liquids and gases
- Particle diagrams and kinetic particle theory
- Changes of state
- Heating and cooling curves
- Gas volume, temperature and pressure
- 1.2 Diffusion and relative molecular mass
- 12 practice MCQs
- 4 structured questions + full answers

Preview

1.1 Solids, liquids and gases

Syllabus 1.1

YOU SHOULD BE ABLE TO

- State the distinguishing properties of solids, liquids and gases.
- Describe the structures of solids, liquids and gases in terms of particle **separation**, **arrangement** and **motion**.
- Describe and explain changes of state (melting, boiling, evaporating, freezing and condensing) using kinetic particle theory.
- Interpret and explain heating and cooling curves using kinetic particle theory.
- Describe and explain the effects of temperature and pressure on the volume of a gas.

The three states of matter are told apart by **shape**, **volume** and **compressibility**. The **kinetic particle theory** explains these properties in terms of the particles' **separation**, **arrangement** and **motion**. Learn the table below word for word.

The kinetic particle theory

All matter is made of tiny **particles** (atoms, molecules or ions) that are **always moving**; the higher the temperature, the greater their average **kinetic energy**. There are **forces of attraction** between particles, and the balance between these forces and the particles' kinetic energy decides the state.

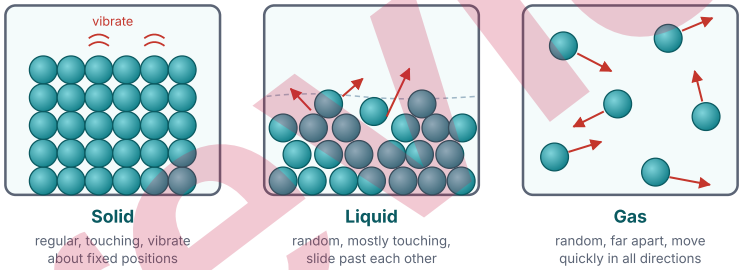


Figure 1.1 Particle diagrams for the three states (red arrows show motion). Liquid particles are still close together.

	Solid	Liquid	Gas
Shape	Fixed shape	Takes the shape of its container	Fills the whole container
Volume	Fixed volume	Fixed volume	No fixed volume
Compressibility	Cannot be compressed	Cannot be compressed	Easily compressed
Separation	Very close, touching	Close, mostly touching	Far apart
Arrangement	Regular (lattice)	Random (irregular)	Random
Motion	Vibrate about fixed positions	Slide past each other	Move quickly and randomly
Forces of attraction	Strong	Weaker, but significant	Very weak (negligible)

KEY POINT — LINKING STRUCTURE TO PROPERTIES

Solid: fixed positions held by strong forces → fixed shape; touching → incompressible. **Liquid**: particles slide past each other → flows; still close → fixed volume. **Gas**: particles move freely → fills container; large spaces between particles → easily compressed.

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COMMON MISTAKE

- Saying solid particles "do not move": they **vibrate**.
- Drawing liquid particles far apart: most should be **touching**, arranged randomly.
- "Particles expand" on heating: the **spaces** grow, not the particles.

EXAM TIP — DRAWING PARTICLE DIAGRAMS

Same-size circles in every box. Solid: 9+ circles in neat rows, all touching. Liquid: random, most touching, no rows. Gas: 5 to 8 circles well spread out, none touching.

Changes of state

A change of state happens when particles gain enough energy to overcome the forces of attraction (or lose so much that the forces hold them together). For a pure substance, freezing point = melting point.

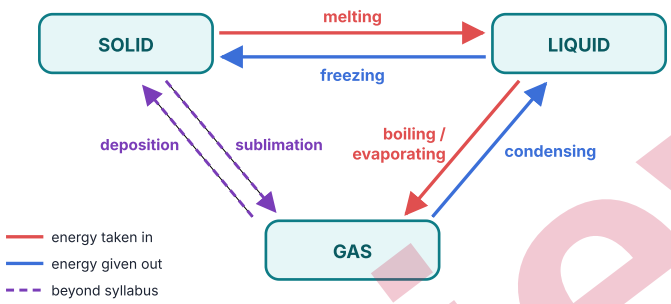


Figure 1.2 The changes of state. Red changes need energy (endothermic); blue changes release energy (exothermic).

Change	What happens to the particles (exam wording)
Melting solid → liquid, at m.p.	Particles gain kinetic energy and **vibrate more**; at the melting point they **overcome the strong forces of attraction** holding them in fixed positions. The lattice breaks down and particles slide past each other.
Boiling liquid → gas, at b.p.	Particles **throughout** the liquid have enough energy to **overcome the forces of attraction completely**; they move far apart as a gas, forming bubbles inside the liquid.
Evaporating liquid → gas, below b.p.	The **most energetic particles at the surface** have enough energy to overcome the attractive forces and escape into the air.
Condensing gas → liquid	On cooling, particles lose kinetic energy and move more slowly; forces of attraction pull them close together. Energy is **given out**.
Freezing liquid → solid, at f.p.	Particles lose kinetic energy; forces of attraction hold them in **fixed positions in a regular arrangement**. Energy is given out.

Evaporation versus boiling

Feature	Evaporation	Boiling
Temperature	Any temperature below the boiling point	Only at the boiling point
Where	Only at the surface of the liquid	Throughout the liquid
Bubbles	No bubbles	Bubbles of gas form inside the liquid
Temperature of liquid	Falls (most energetic particles leave)	Constant while boiling

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BEYOND THE SYLLABUS — EVAPORATION, PRESSURE AND SUBLIMATION

- **Evaporation cools** the liquid left behind: the fastest particles escape, so the average kinetic energy falls (sweat cools you).
- **Sublimation**: solid → gas directly, without melting (dry ice, solid CO_2; iodine gives a purple vapour). The reverse is **deposition**.

Using melting and boiling points to predict state

Compare the temperature with the m.p. and b.p.: **below m.p. → solid**; **between m.p. and b.p. → liquid**; **above b.p. → gas**. Watch negative numbers: −50 °C is *colder* than −10 °C, so sketch a quick number line.

WORKED EXAMPLE — PREDICTING STATES

Substance	m.p. / °C	b.p. / °C
P	−101	−35
Q	−117	78
R	44	280

Q. State the physical state of each substance at 25 °C.

P: 25 °C is above its b.p. (−35 °C) → **gas**.

Q: 25 °C lies between −117 °C and 78 °C → **liquid**.

R: 25 °C is below its m.p. (44 °C) → **solid**.

(At −40 °C, P would be a liquid: below −35 °C but above −101 °C.)

Purity: a **pure** substance melts and boils at **sharp, fixed** temperatures (pure water: 0 °C and 100 °C). An impurity **lowers the melting point** (melting over a range) and **raises the boiling point** (see Chapters 10 and 12).

Heating curves

Heat a solid at a steady rate, record its temperature regularly and plot temperature against time: this is a **heating curve**. Sloping sections (temperature rising) alternate with **flat plateaus** where the state is changing.

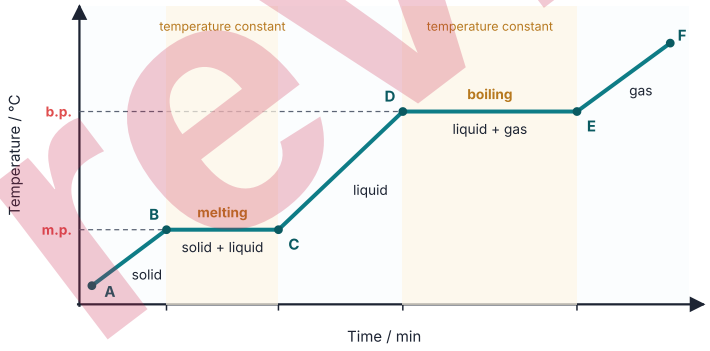


Figure 1.3 Heating curve of a pure substance. Shaded plateaus are changes of state: energy is used to overcome forces of attraction, not to raise the temperature.

Sloping sections: AB, CD, EF

One state only (AB solid, CD liquid, EF gas). Energy **increases the kinetic energy** of the particles (they vibrate or move faster), so the **temperature rises**.

Plateaus: BC (melting), DE (boiling)

Two states present. Energy is used to **overcome the forces of attraction** between particles, **not to increase their kinetic energy**, so the **temperature stays constant**.

KEY POINT — WHY IS DE LONGER THAN BC?

Boiling needs **much more energy** than melting: forces of attraction must be overcome **completely**, not just loosened (water: 6 kJ/mol to melt, 41 kJ/mol to boil).

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Cooling curves

A **cooling curve** is the reverse: a molten substance is left to cool and its temperature recorded at regular intervals, a common Paper 3/4 experiment (e.g. stearic acid, m.p. 69 °C, melted in a water bath).

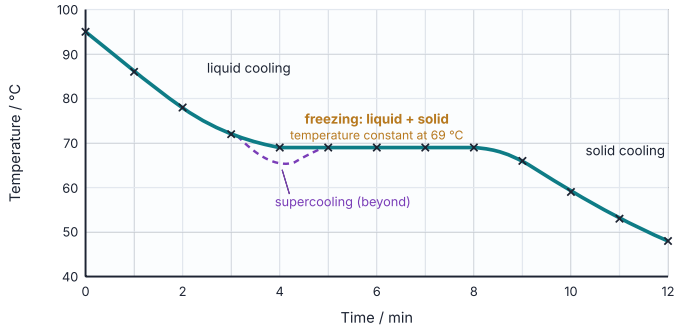


Figure 1.4 Cooling curve of stearic acid from 95 °C (crosses are the plotted readings; draw a smooth best-fit line). The plateau gives the freezing point, 69 °C.

EXAM TIP — THE PLATEAU EXPLANATION

Heating plateau: "energy is used to **overcome the forces of attraction** between particles, **not to increase their kinetic energy**." **Cooling** plateau: "energy is **released as forces of attraction form** between particles; this balances the energy lost to the surroundings, so the temperature stays the same." Never write that the particles "stop gaining energy". Read the m.p./f.p. from the **flat part**; a sloping "plateau" suggests an **impure** sample.

BEYOND THE SYLLABUS — SUPERCOOLING

A clean, still liquid may cool slightly **below** its f.p. before crystals form (dashed dip in Figure 1.4); the energy released then brings it back up to the f.p. Read the flat part, not the dip.

Gases: the effect of temperature and pressure on volume

Gas particles constantly hit the walls of their container; these collisions produce the **gas pressure**. Remember: **"hot gas grows, squeezed gas shrinks"**, and always use the phrase "**more frequent collisions with the walls**".

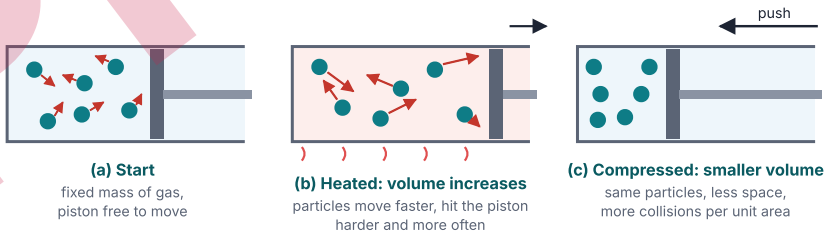


Figure 1.5 A fixed mass of gas in a cylinder with a free piston. (b) Raising the temperature makes the gas expand. (c) Increasing the pressure squashes the gas into a smaller volume.

Temperature up (constant pressure)

Particles gain kinetic energy and **move faster**, hitting the walls **more frequently** and **with more force**, so the gas pushes the piston out: the **volume increases**. (In a rigid container the **pressure increases** instead.)

Pressure up (constant temperature)

The same number of particles is forced into a smaller space: they move **closer together**, possible because gas particles are **far apart**. The **volume decreases** and the particles hit the walls more frequently.

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BEYOND THE SYLLABUS — THE GAS LAWS AND THE KELVIN SCALE

For a fixed mass of gas the relationships are exact if temperature is in **kelvin**: **T / K = θ / °C + 273** (0 K = −273 °C, **absolute zero**).

- **Boyle's law** (constant T): **pV = constant**, so $p_1V_1 = p_2V_2$. E.g. 240 cm³ at 100 kPa compressed to 300 kPa: $V_2 = 100 \times 240/300 =$ **80 cm³**.
- **Charles' law** (constant p): **V / T = constant**, so $V_1/T_1 = V_2/T_2$. E.g. 150 cm³ at 27 °C heated to 127 °C: $T_1 = 300$ K, $T_2 = 400$ K, so $V_2 = 150 \times 400/300 =$ **200 cm³** (using °C would wrongly give 706 cm³).

1.2 Diffusion

Syllabus 1.2

YOU SHOULD BE ABLE TO

- Describe and explain diffusion in terms of kinetic particle theory.
- Describe and explain the effect of relative molecular mass (M_r) on the rate of diffusion of gases.

DEFINITION

Diffusion is the **net movement of particles from a region of higher concentration to a region of lower concentration**, as a result of their **random motion**. It continues until the particles are evenly spread out.

Particles of a gas or liquid move **randomly** and collide. Where there are more particles, more happen to move *away* than move in, giving a **net** spread from high to low concentration, with no stirring needed. Once evenly mixed there is **no net movement**.

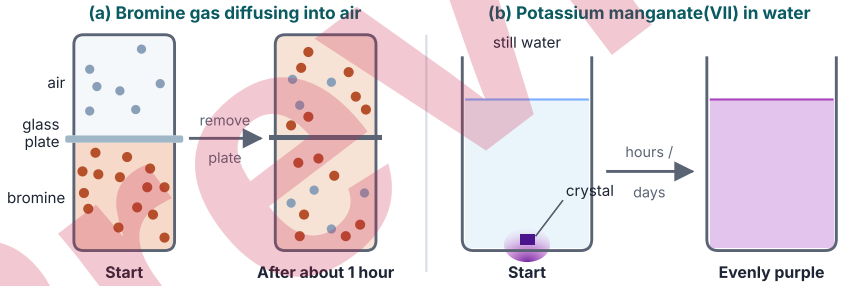


Figure 1.6 Diffusion in a gas and in a liquid. (a) Red-brown bromine spreads upwards into air even though it is denser than air. (b) The purple colour spreads slowly through still water.

Gases diffuse fast (bromine fills two gas jars in about an hour): their particles move quickly and are far apart. **Liquids diffuse slowly** (hours to days for the purple colour): particles move more slowly and are close together, so they collide very frequently. Diffusion in **solids** is negligible, as particles only vibrate.

Effect of relative molecular mass

At the same temperature, all gas particles have the same average kinetic energy, so **lighter particles move faster**: **the lower the M_r, the faster the gas diffuses**.

Gas (fastest → slowest)	H_2	He	CH_4	NH_3	N_2	O_2	HCl	CO_2	Cl_2	Br_2
M_r	2	4	16	17	28	32	36.5	44	71	160

The ammonia and hydrogen chloride experiment

Cotton wool soaked in **concentrated ammonia solution** and cotton wool soaked in **concentrated hydrochloric acid** are placed **at the same moment** at opposite ends of a long, dry, horizontal glass tube (in a fume

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cupboard), and both ends are sealed. The gases diffuse along the tube; where they meet, a **white ring** of solid forms after a few minutes.

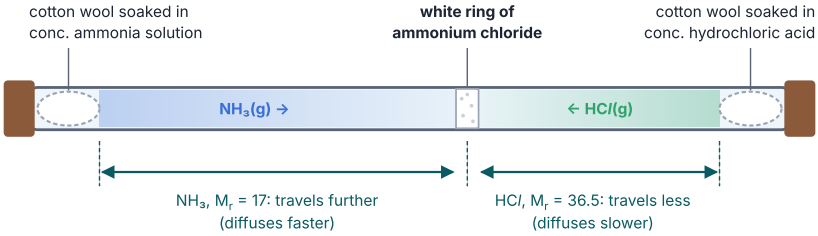


Figure 1.7 The white ring of ammonium chloride forms nearer the hydrochloric acid end, because ammonia diffuses faster.

KEY POINT — THE MODEL ANSWER

$$NH_3(g) + HCl(g) \rightarrow NH_4Cl(s)$$

"The white ring forms nearer the HCl end. NH_3 has a **lower M_r** (17) than HCl (36.5), so NH_3 molecules **move faster** and **diffuse faster**; in the same time NH_3 travels further along the tube than HCl." With HBr (M_r 81) in place of HCl, the ring forms **even closer** to the acid end.

BEYOND THE SYLLABUS — GRAHAM'S LAW

Rate of diffusion $\propto 1/\sqrt{M_r}$, so rate(NH_3)/rate(HCl) = $\sqrt{(36.5/17)}$ = 1.47. In a 100 cm tube the ring is predicted 100 × 1.47/2.47 = **59 cm from the ammonia end**.

Effect of temperature on the rate of diffusion

At a **higher temperature** particles have **more kinetic energy** and **move faster**, so diffusion is faster: $KMnO_4$ spreads more quickly in hot water; the white ring forms sooner in a warm room (same position).

BEYOND THE SYLLABUS — BROWNIAN MOTION

Smoke particles in air (seen under a microscope) move in a **random, zigzag path** because they are hit **unevenly** by invisible, fast-moving air molecules: evidence of constant random motion.

Chapter summary

- Solids: fixed shape and volume. Liquids: fixed volume, no fixed shape. Gases: neither; easily compressed.
- Solid: touching, regular, vibrate. Liquid: close, random, slide past each other. Gas: far apart, random, fast in all directions.
- Melting, boiling, evaporating take in energy to overcome forces of attraction; condensing and freezing give energy out.
- Evaporation: surface only, below b.p., no bubbles. Boiling: throughout, at b.p., bubbles. Below m.p. solid; between m.p. and b.p. liquid; above b.p. gas.
- Plateaus = change of state at constant temperature: energy overcomes forces (heating) or is released as forces form (cooling). Boiling plateau is longest.
- Gas heated → particles faster → volume up (pressure up if rigid). Compressed → particles closer → volume down.
- Diffusion: net movement from higher to lower concentration by random motion; fast in gases, slow in liquids.
- Lower M_r or higher temperature → faster particles → faster diffusion. NH_4Cl ring forms nearer the HCl end (17 vs 36.5).

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Practice — Chapter 1

Multiple-choice (Paper 1 style) and structured questions (Paper 2 and Paper 4 style). Answers at the end.

Multiple-choice questions

1 Which description of the particles in a liquid is correct?

A regular arrangement, close together, vibrate about fixed positions

B random arrangement, far apart, move quickly in all directions

C regular arrangement, close together, slide past each other

D random arrangement, close together, slide past each other

2 The melting points and boiling points of four substances are shown. Which substance is a liquid at 80 °C?

Substance	m.p. / °C	b.p. / °C
A	−210	−196
B	−7	59
C	0	100
D	114	184

3 Why does the temperature of a pure solid stay constant while it is melting?

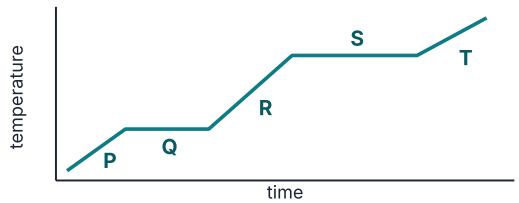
A Energy is used to overcome the forces of attraction between the particles.

B The particles stop moving.

C The kinetic energy of the particles increases.

D The solid stops absorbing heat.

4 The graph shows the heating curve of a pure substance. In which section are the particles changing from a regular arrangement to a random arrangement?



A P **B** Q **C** R **D** S

6 A sealed gas syringe with a freely moving plunger is placed in hot water. Which row is correct?

	speed of gas particles	volume of gas
A	increases	increases
B	increases	decreases
C	no change	increases
D	decreases	increases

7 The plunger of a sealed syringe containing a gas is pushed in at constant temperature. Why does the pressure of the gas increase?

A The gas particles become smaller.

B The gas particles move more slowly.

C The gas particles collide with the walls more frequently.

D The number of gas particles increases.

8 Which gas diffuses fastest at room temperature?

A chlorine, Cl_2 **B** carbon dioxide, CO_2

C nitrogen, N_2 **D** methane, CH_4

9 The ammonia and hydrogen chloride diffusion experiment is repeated using concentrated hydrobromic acid (HBr, M_r = 81) instead of hydrochloric acid. Where does the white ring form?

A nearer the ammonia end than before

B exactly halfway along the tube

C nearer the acid end than before

D in the same place as before

10 A crystal of potassium manganate(VII) is placed in a beaker of still water. Which change makes the purple colour spread faster?

A cooling the water **B** warming the water

C covering the beaker **D** using a taller beaker

Preview

5 Which statement about evaporation is correct?

A It occurs only at the boiling point.
B It takes place throughout the liquid.
C Bubbles of gas form inside the liquid.
D It takes place only at the surface of the liquid.

11 Which change of state releases energy to the surroundings?

A condensing **B** boiling
C evaporating **D** melting

12 Three statements about the particles in a gas are given.
1 They are far apart. 2 They move randomly. 3 There are strong forces of attraction between them.
Which statements are correct?

A 1 only **B** 1 and 2 only
C 2 and 3 only **D** 1, 2 and 3

Structured questions

Question 1

The table shows the melting points and boiling points of four substances.

Substance	m.p. / °C	b.p. / °C
ethanol	−114	78
mercury	−39	357
oxygen	−218	−183
sodium chloride	801	1413

(a) Which substance is a gas at 25 °C? [1]

(b) Which two substances are liquids at 25 °C? [1]

(c) State the physical state of ethanol at −120 °C. [1]

(d) Describe the arrangement and motion of the particles in solid sodium chloride. [2]

(e) Mercury is heated from 25 °C to 400 °C. Describe the changes in the movement and separation of the mercury particles. [3]

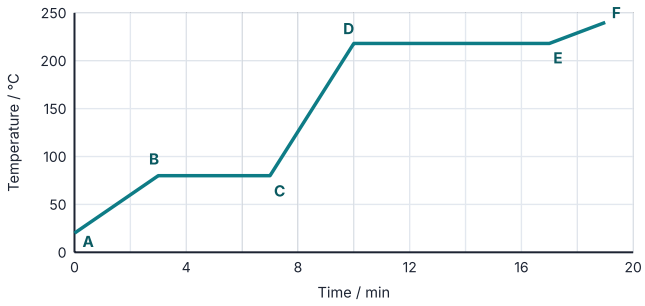
(f) Explain why oxygen gas can be compressed easily but liquid ethanol cannot. [2]

[Total: 10]

Preview

Question 2

Solid Y is heated at a steady rate. The graph shows how its temperature changes.



(a) Use the graph to give the melting point and the boiling point of Y. [2]

(b) State the physical state(s) of Y (i) in section BC, (ii) at 150 °C. [2]

(c) Use kinetic particle theory to explain why the temperature stays constant in section BC. [2]

(d) Suggest why section DE is longer than section BC. [2]

(e) Liquid Y also evaporates slowly at 100 °C. Give two differences between evaporation and boiling. [2]

(f) Gaseous Y is cooled until it turns into a liquid. Name this change of state and state whether energy is taken in or given out. [2]

[Total: 12]

Question 3

(a) A sealed gas syringe with a freely moving plunger contains 50 cm^3 of air at 20 °C. It is placed in water at 80 °C.

(i) State what happens to the volume of the air. [1]

(ii) Explain your answer in terms of the particles. [3]

(b) The plunger is then pushed in at constant temperature. Explain why the pressure of the air increases. [2]

(c) A student sets up the apparatus below. After a few minutes a white ring forms in the tube.



(i) Name the white solid. [1]

(ii) Write the equation, with state symbols, for its formation. [2]

(iii) Explain why the white ring forms nearer the hydrochloric acid end. Refer to relative molecular masses in your answer. [3]

(iv) The experiment is repeated in a warmer laboratory. State and explain the effect on the time taken for the ring to form. [2]

[Total: 14]

Question 4 (Paper 4 style)

A student melts solid X in a boiling tube held in a beaker of hot water. The tube is then removed and the temperature of X is recorded every half minute as it cools.

time / min	0	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0	5.5	6.0
temperature / °C	95	90	86	83	80	80	80	80	80	78	70	72	69

Preview